

THE ANATOMY OF A PHONE NUMBER

Anatomy of a 5G Voice Call

VOLUME II OF III

From pressing CALL to a distributed transaction across a dozen protocols

A single 5G Standalone voice call, followed end to end: cell search, registration, authentication, PDU sessions, IMS registration, the SIP INVITE, inter-operator routing, paging, codec negotiation, QoS and the radio stack that finally carries the speech.



A LAYERED STUDY OF TELEPHONE NUMBERING, MOBILE NETWORKS AND PACKET TELEPHONY

PART II / III

About this volume

A single 5G Standalone voice call, followed end to end: cell search, registration, authentication, PDU sessions, IMS registration, the SIP INVITE, inter-operator routing, paging, codec negotiation, QoS and the radio stack that finally carries the speech.

This is one of three volumes. Each descends a level further from the digits a human types toward the electromagnetic wave that eventually carries the speech. The volumes are meant to be read in order, but each is self-contained.

VOLUME I

The Architecture of a Phone Number

VOLUME II

Anatomy of a 5G Voice Call

VOLUME III

Packet-Level Telecommunications

Diagrams are numbered by chapter. A list of figures follows the contents, so any diagram can be found directly.

Contents

CH 1	Setting the Scene	8
1.1	The scenario	8
1.2	First correction: the phone doesn't start by asking the Internet	9
1.3	The complete architecture	10
CH 2	Attachment: SIM, Cell Search and Registration	13
2.1	Before the call: SIM and subscriber identity	13
2.2	Step zero: the phone finds a cell	14
2.3	Registration with the 5G Core	15
2.4	AMF	16
CH 3	Authentication and the Security Context	17
3.1	Authentication	17
3.2	Security context	18
3.3	Registration complete	18

CH 4	PDU Sessions and IP Connectivity	20
4.1	Establishing a PDU session	20
4.2	The phone gets IP connectivity	21
CH 5	IMS Registration	22
5.1	IMS registration	22
5.2	P-CSCF	22
5.3	SIP REGISTER	23
5.4	HSS/UDM enters the picture	24
5.5	S-CSCF	24
5.6	IMS registration completes	25
CH 6	Placing the Call: the SIP INVITE	26
6.1	Now Alice presses CALL	26
6.2	SIP INVITE	27
6.3	P-CSCF receives the INVITE	27
6.4	S-CSCF analyzes the destination	28
CH 7	Inter-Operator Routing	29
7.1	If Bob is on another operator	29
7.2	I-CSCF	30
7.3	Bob's S-CSCF	31
7.4	TAS / Application Server	32
CH 8	Finding Bob: Paging and Alerting	34
8.1	Bob is currently idle	34
8.2	Paging is not "call every tower"	35
8.3	Bob's phone responds	36
8.4	SIP alerting	36
8.5	Bob answers	37
CH 9	Media: Codecs, QoS and RTP	39
9.1	But we still don't have the conversation	39
9.2	Codec negotiation	39
9.3	Now QoS becomes important	40
9.4	The voice packet path	41

9.5 RTP	42
9.6 One second of speech	43
CH 10 The 5G Radio Stack	45
10.1 The actual 5G radio stack	45
10.2 PHY	46
10.3 MAC	46
10.4 RLC	47
10.5 PDCP	47
10.6 gNB	47
10.7 N2 and N3	48
10.8 UPF	48
CH 11 Generations Compared	50
11.1 What happened to the old MSC?	50
11.2 LTE VoLTE	52
11.3 5G VoNR	52
11.4 What if the destination is a landline?	53
11.5 What happens when Alice hangs up?	54
CH 12 The Entire Call in One Sequence	56
12.1 The entire call in one sequence	56
CH 13 Identifiers, Mobility and Roaming	58
13.1 The really interesting part: there are multiple identifiers	58
13.2 This is exactly like a computer network	60
13.3 Why the number doesn't need to know your location	60
13.4 And now you can understand roaming	62
13.5 Where SS7 fits now	63
CH 14 Protocol Roles and the Deepest Model	65
14.1 What each protocol is doing	65
14.2 The deepest mental model	66
14.3 The entire thing from digit to electromagnetic wave	67

Figures

Fig. 1.1	Alice / Bob → Alice's UE / Bob's UE → Alice's operator / Bob's operator	8
Fig. 1.2	Phone → "where is +91 98BBBBBB20?"	9
Fig. 1.3	Before call → Ready state → Call procedure	10
Fig. 1.4	Other operator → AMF / → UE /	11
Fig. 1.5	Control plane → User plane → destination	12
Fig. 2.1	Phone number → secure hardware	14
Fig. 2.2	Structure of Phone	14
Fig. 2.3	Phone → reads system information	15
Fig. 2.4	UE → AMF	15
Fig. 2.5	UE → AMF	16
Fig. 3.1	UE → AUSF → subscriber authentication data	17
Fig. 3.2	Ue / network → derive security context /	18
Fig. 3.3	UE and its constituent parts	19
Fig. 4.1	UE → SMF → data network	20
Fig. 4.2	UE → IMS	21
Fig. 5.1	UE → IMS	23
Fig. 5.2	UE → S-CSCF	23
Fig. 5.3	subscriber → current registration/contact	24
Fig. 5.4	MSISDN → S-CSCF → Bob's UE	25
Fig. 6.1	Dialer → Sip invite	26
Fig. 6.2	Alice UE → P-CSCF	28
Fig. 6.3	S-CSCF → "Where does this number belong?"	28
Fig. 7.1	Alice Operator → Bob Operator	29
Fig. 7.2	Alice S-CSCF → Bob S-CSCF	29
Fig. 7.3	Alice's network → route to Bob's home/serving IMS	30
Fig. 7.4	I-CSCF → obtain current S-CSCF → for a terminating subscriber. (3GPP)	31
Fig. 7.5	Alice → interconnect → Bob S-CSCF	31
Fig. 7.6	number → service profile → destination	33
Fig. 8.1	Incoming call → AMF / mobility state → Bob's UE	35
Fig. 8.2	Tracking Area 17 → gNB A / gNB B / gNB C	35
Fig. 8.3	AMF → gNBs → Bob's UE	36

Fig. 8.4	Alice / Bob → 180 Ringing / → phone rings /	37
Fig. 8.5	Bob → Alice	38
Fig. 9.1	analog-ish speech → compressed speech frames	40
Fig. 9.2	MIC → voice processing → compressed frame	40
Fig. 9.3	voice packet → retransmit in 800 ms	41
Fig. 9.4	sound → UPF → Bob's ear	42
Fig. 9.5	RTP packet → compressed voice payload	43
Fig. 9.6	speech → packetized → reassembled/played	44
Fig. 10.1	Application → RLC → radio waveform	45
Fig. 10.2	bits → RF → electromagnetic wave	46
Fig. 10.3	gNB → user plane	48
Fig. 10.4	CONTROL → UPF → external network	49
Fig. 11.1	UE → BSC/RNC → PSTN	50
Fig. 11.2	UE → EPC → with VoLTE	51
Fig. 11.3	UE → 5GC → with VoNR where deployed	51
Fig. 11.4	UE → IMS	52
Fig. 11.5	UE → IMS	53
Fig. 11.6	IMS → traditional telephone	54
Fig. 11.7	1890s telephone → 2026 smartphone	54
Fig. 11.8	Alice → Bob	55
Fig. 12.1	Before call → I-CSCF → resources released	56
Fig. 13.1	MSISDN → IP address → These identities exist at different layers	59
Fig. 13.2	Delhi → London	61
Fig. 13.3	STATIC-ish → DYNAMIC → physical transmission	62
Fig. 13.4	Home network → Germany → UE	63
Fig. 13.5	Telephone signaling → Diameter → HTTP/2-based service APIs	64
Fig. 14.1	Identifier resolution → Resource control → PHYSICAL	66
Fig. 14.2	Control plane → what resources, what state?" → "Here are the actual bits."	67
Fig. 14.3	+91 98bbbbbb20 → RF waveform → Bob's eardrum	68

Introduction

Now we get into the machinery. This is where the innocent act of tapping Call turns into a distributed transaction spanning the handset, radio access network, 5G core, subscriber databases, IMS, SIP signaling, QoS policy, and finally an RTP media stream. Humanity somehow made talking to another human require an entire civilization's worth of infrastructure.

I'll use a 5G Standalone (5G SA) → VoNR call as the primary example, then show where LTE/VoLTE and legacy 2G/3G differ.

3GPP's current 5G system procedures are specified primarily in TS 23.502, while IMS architecture is specified in TS 23.228 and SIP/SDP call control in TS 24.229. These specifications remain under change control.

CHAPTER 1

Setting the Scene

1.1 The scenario

1.3 The complete architecture

1.2 First correction: the phone doesn't start by asking the Internet

1.1 The scenario

Let's define two subscribers:

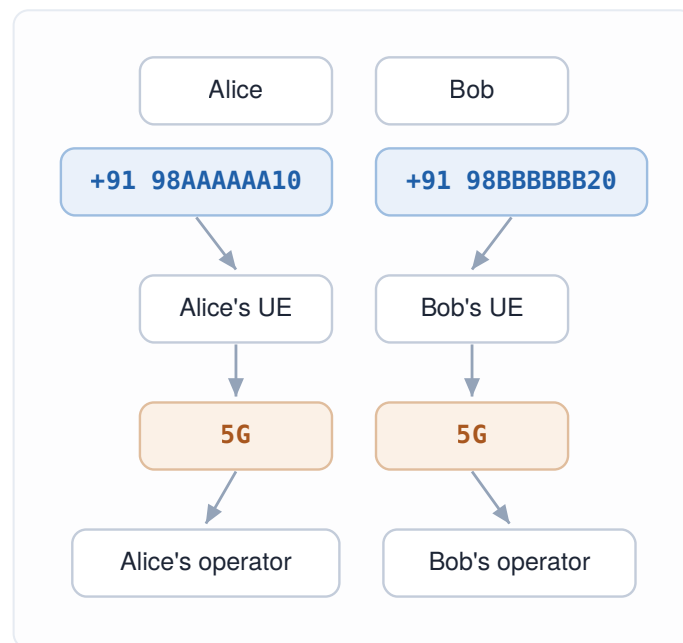


Figure 1.1 Alice / Bob → Alice's UE / Bob's UE → Alice's operator / Bob's operator

Alice presses:

CALL

and enters:

+91 98BBBBBB20

Our objective is to follow the call all the way down.

1.2 First correction: the phone doesn't start by asking the Internet

Your phone does not do:



Figure 1.2 Phone → "where is +91 98BBBBBB20?"

Nor does the phone send the entire telephone number over the radio and wait for a tower to figure everything out.

There is already a substantial amount of state established before you press Call.

The phone has previously:

- registered with the mobile network
- authenticated
- established security
- obtained network configuration
- discovered IMS services
- registered to IMS

So the call is really:

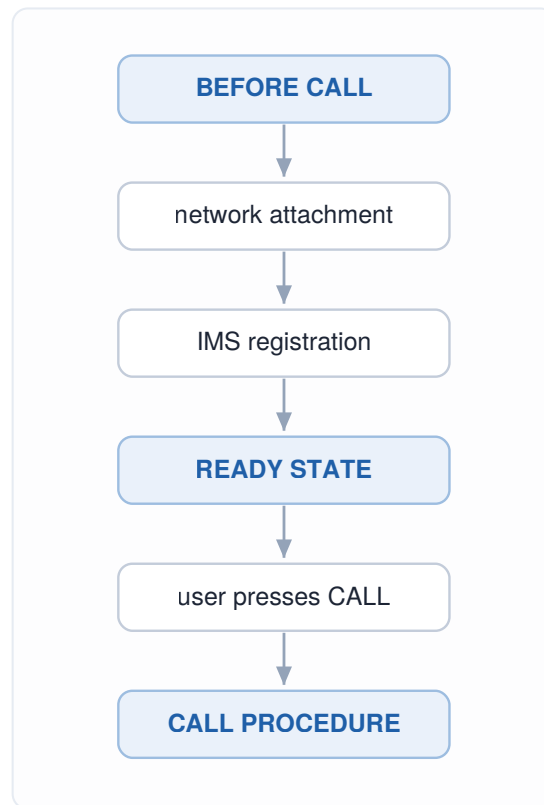


Figure 1.3 Before call → Ready state → Call procedure

This distinction is critical.

1.3 The complete architecture

A simplified 5G SA voice architecture looks like this:

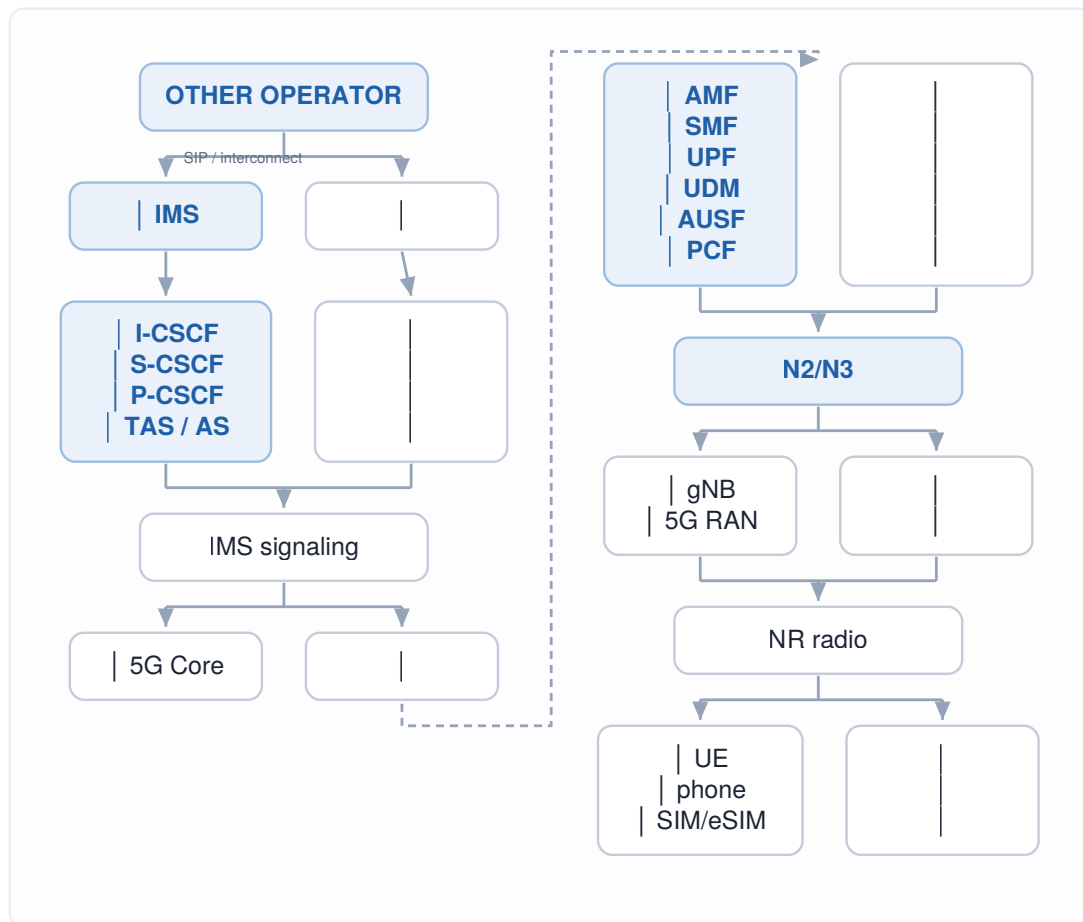


Figure 1.4 Other operator → | AMF / | → | UE / |

There are actually two major planes involved:

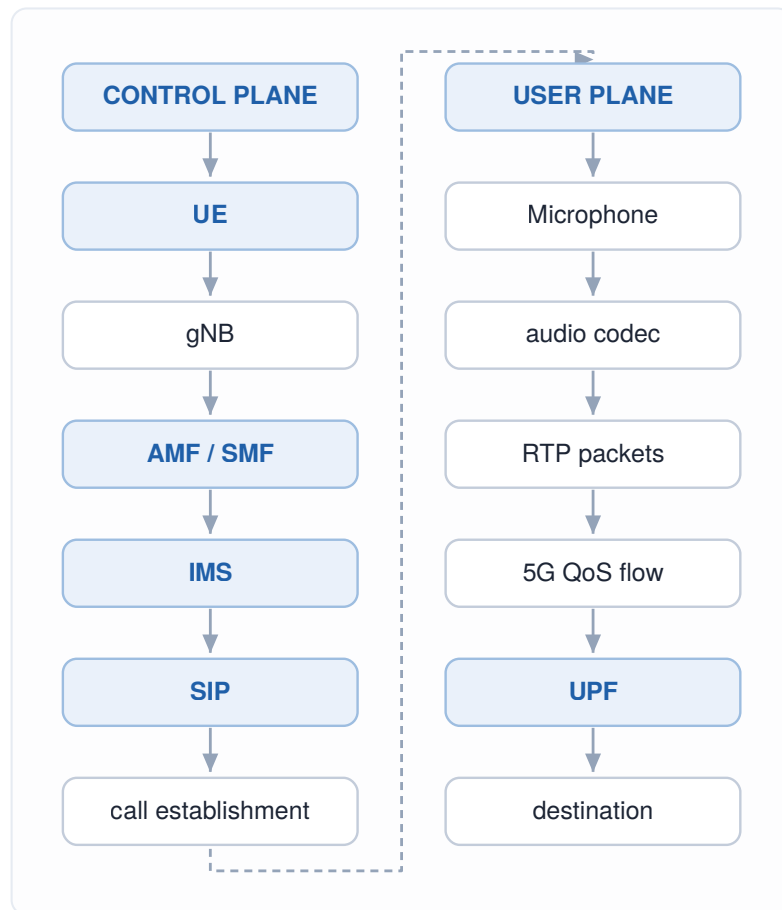


Figure 1.5 Control plane → User plane → destination

The call's signaling and its audio are therefore fundamentally different traffic.

Attachment: SIM, Cell Search and Registration

[2.1 Before the call: SIM and subscriber identity](#)

[2.3 Registration with the 5G Core](#)

[2.2 Step zero: the phone finds a cell](#)

[2.4 AMF](#)

2.1 Before the call: SIM and subscriber identity

Let's start much earlier.

Your SIM/eSIM contains subscriber credentials and identifiers.

Historically this is represented around:

IMSI

and in 5G the permanent subscription identity is conceptually represented as:

SUPI

with the network normally avoiding exposing the permanent identity over the air unnecessarily.

A simplified identity stack:

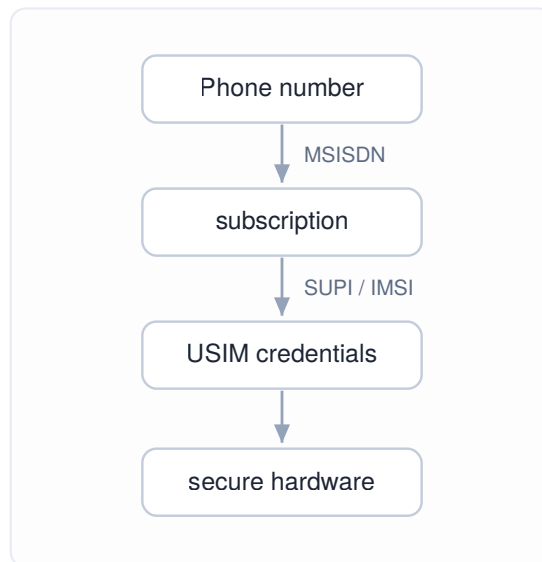


Figure 2.1 Phone number → secure hardware

Separately:

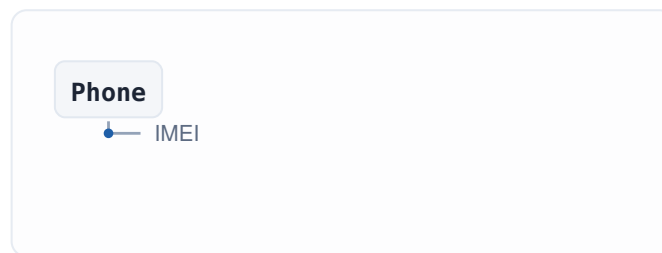


Figure 2.2 Structure of Phone

Remember:

- MSISDN** → telephone number
- SUPI/IMSI** → subscription identity
- IMEI** → equipment identity

These are different namespaces.

2.2 Step zero: the phone finds a cell

Before doing almost anything else, the UE has to discover the radio network.

Conceptually:

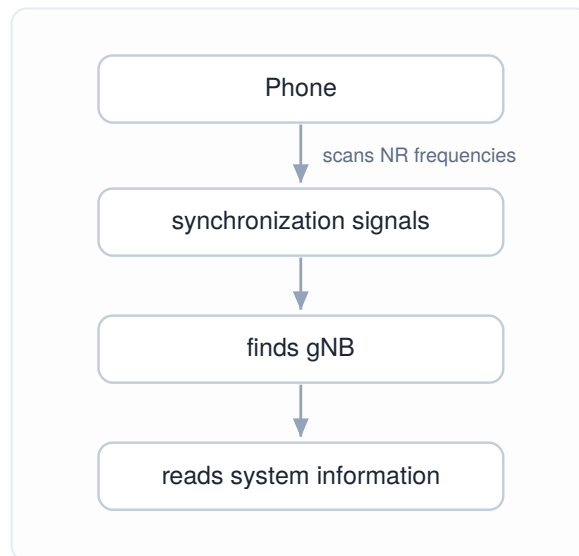


Figure 2.3 Phone → reads system information

learns:

- - network identity
- - access parameters
- - cell information
- - configuration

The phone now knows:

"There is a 5G network here, and this is how I interact with it."

The gNB is the 5G base station.

2.3 Registration with the 5G Core

The phone performs 5G registration.

The radio path is approximately:

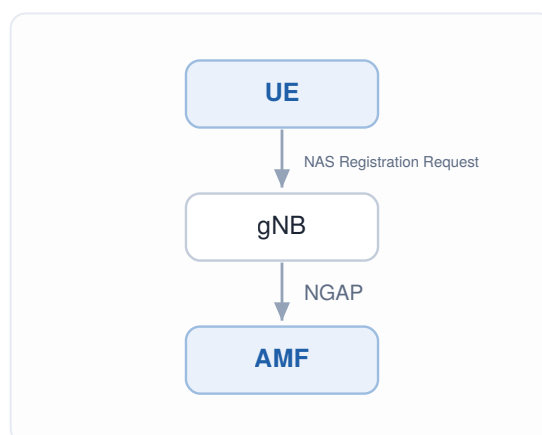


Figure 2.4 UE → AMF

Notice something subtle:

NAS

doesn't directly travel through the radio as some magical independent network.

It is carried through the RAN using lower-layer protocols.

Conceptually:

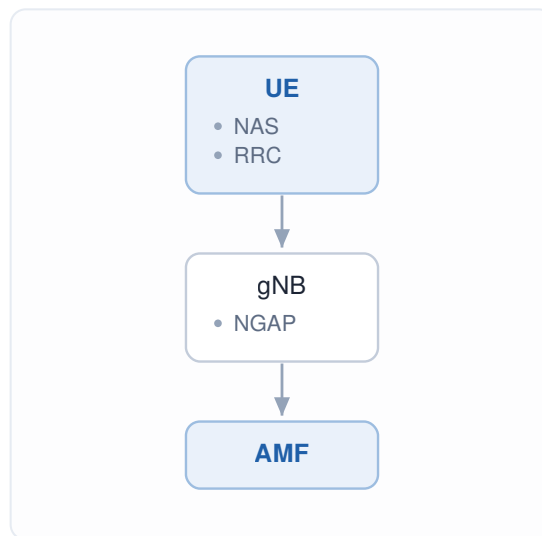


Figure 2.5 UE → AMF

2.4 AMF

The AMF, Access and Mobility Management Function, is one of the central control-plane components of the 5G Core.

Its responsibilities include things such as:

- registration
- connection management
- mobility management
- reachability
- paging
- NAS signaling
- access authentication coordination

AMF

"Where is this UE and what is its control-plane state?"

Not literally only that, but that's a useful mental model.

Authentication and the Security Context

3.1 Authentication

3.2 Security context

3.3 Registration complete

3.1 Authentication

The network needs to establish that the subscriber possesses valid credentials.

The simplified architecture is:

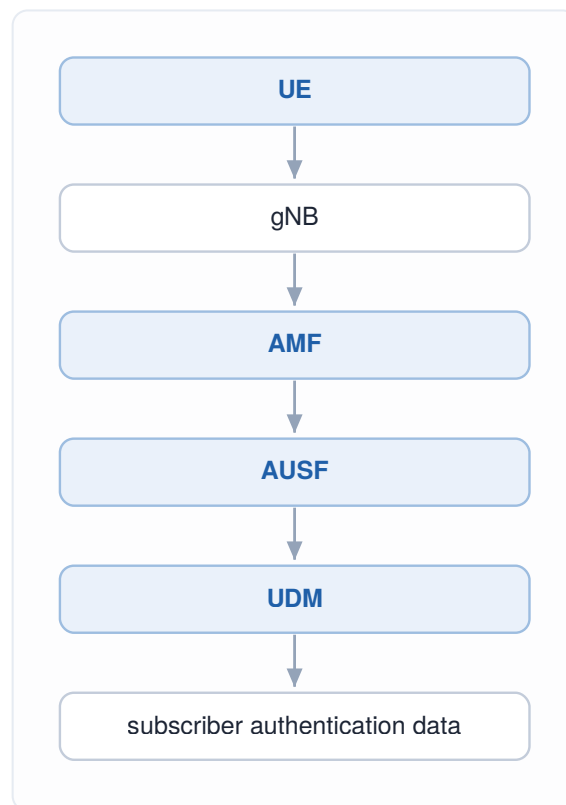


Figure 3.1 UE → AUSF → subscriber authentication data

The important components are:

- AUSF
- Authentication Server Function
- UDM
- Unified Data Management

The UDM is the modern 5G evolution of some functionality historically associated with the HLR/HSS ecosystem.

The authentication procedure uses cryptographic mechanisms associated with the subscriber's credentials.

The network isn't supposed to receive:

"here is my password: 123456"

like some cursed 1990s web application.

Instead, the subscriber and network participate in a standardized authentication procedure.

3.2 Security context

After successful authentication, security is established.

Conceptually:

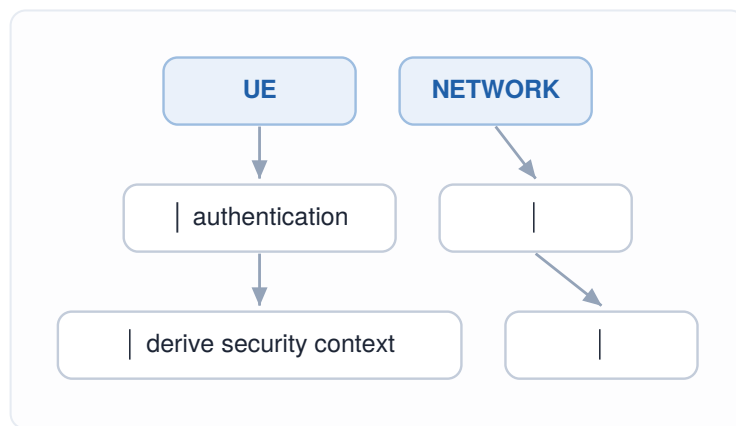


Figure 3.2 Ue / network → | derive security context / |

Now subsequent signaling can be protected.

This gives us:

- identity
- integrity
- authentication
- confidentiality

subject to the specific protocol and network configuration.

3.3 Registration complete

Eventually the UE reaches a state approximately like:

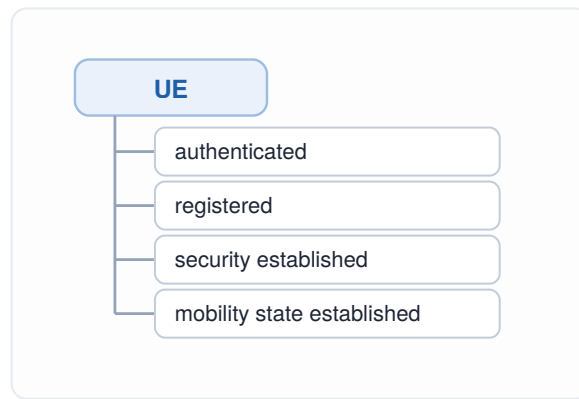


Figure 3.3 UE and its constituent parts

But we're still not ready for VoNR.

We need packet connectivity.

PDU Sessions and IP Connectivity

4.1 Establishing a PDU session

4.2 The phone gets IP connectivity

4.1 Establishing a PDU session

5G doesn't create a dedicated old-school voice circuit.

Instead, the UE establishes packet connectivity.

Conceptually:

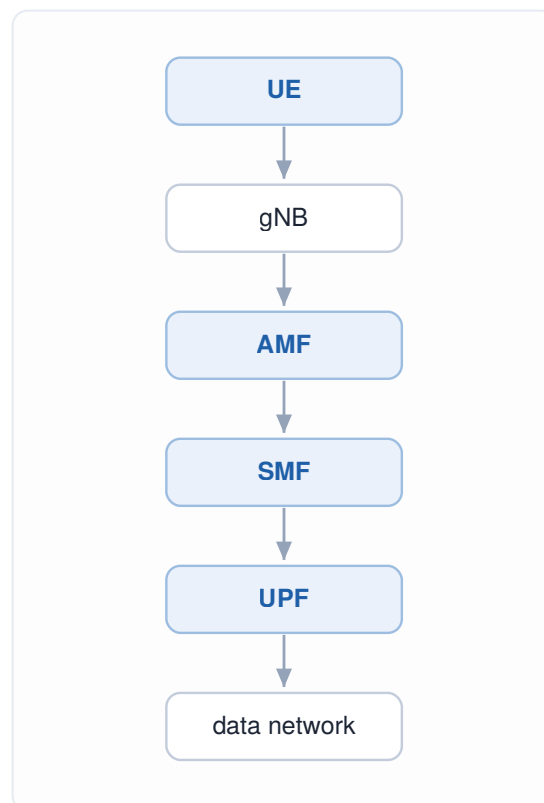


Figure 4.1 UE → SMF → data network

The SMF, Session Management Function, manages the PDU session.

The UPF, User Plane Function, forwards actual user traffic.

- | | | |
|--------------------|----------------------|---------------------|
| • AMF | • SMF | • UPF |
| • mobility/control | • session management | • packet forwarding |

This separation is fundamental to 5G's architecture.

4.2 The phone gets IP connectivity

The PDU session can provide IP connectivity.

Conceptually:

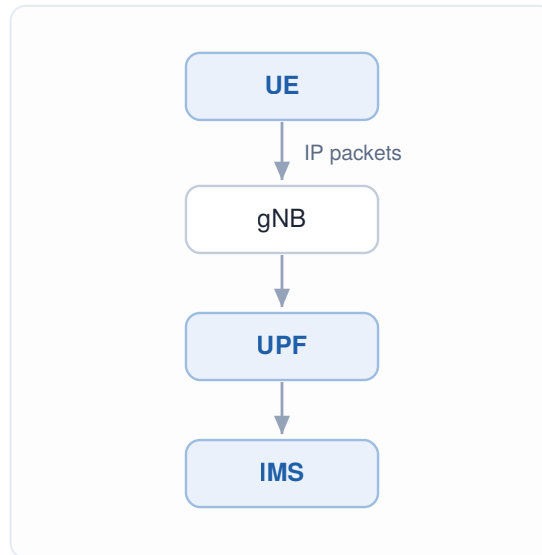


Figure 4.2 UE → IMS

The phone can now communicate with the operator's IMS infrastructure.

And now we reach the important part.

CHAPTER 5

IMS Registration

5.1 IMS registration

5.2 P-CSCF

5.3 SIP REGISTER

5.4 HSS/UDM enters the picture

5.5 S-CSCF

5.6 IMS registration completes

5.1 IMS registration

IMS means:

IP Multimedia Subsystem

3GPP defines IMS architecture around CSCF functions including:

- P-CSCF
- I-CSCF
- S-CSCF
- and application servers. [3GPP Portal](#)

Think of IMS as the telecommunications service-control layer sitting above IP connectivity.

Your 5G network gives you:

IP connectivity

IMS gives you:

- voice services
- video
- messaging
- supplementary services
- session control

5.2 P-CSCF

The phone first communicates with a:

P-CSCF

Proxy-Call Session Control Function.

Conceptually:

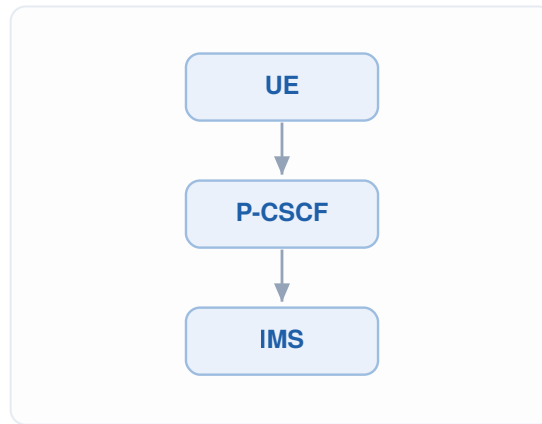


Figure 5.1 UE → IMS

The P-CSCF is the UE's first IMS signaling point.

3GPP's IMS architecture explicitly describes the P-CSCF as the proxy associated with the UE and as part of the SIP signaling path. (3GPP)

5.3 SIP REGISTER

The UE sends a SIP registration message.

Conceptually:

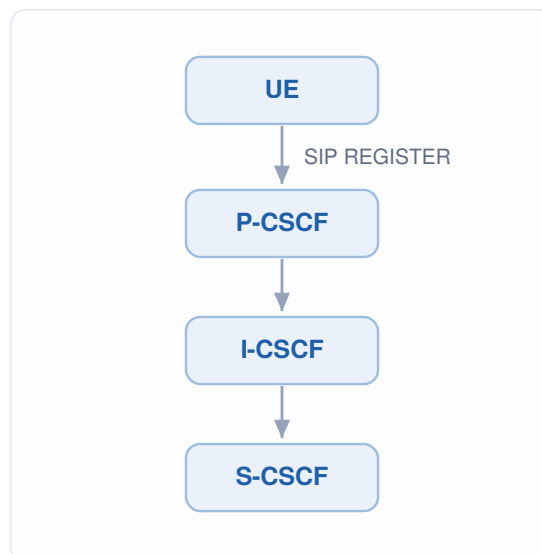


Figure 5.2 UE → S-CSCF

The registration effectively says:

"This IMS subscriber is currently reachable through this IMS registration."

The number isn't enough.

The IMS network needs to associate:

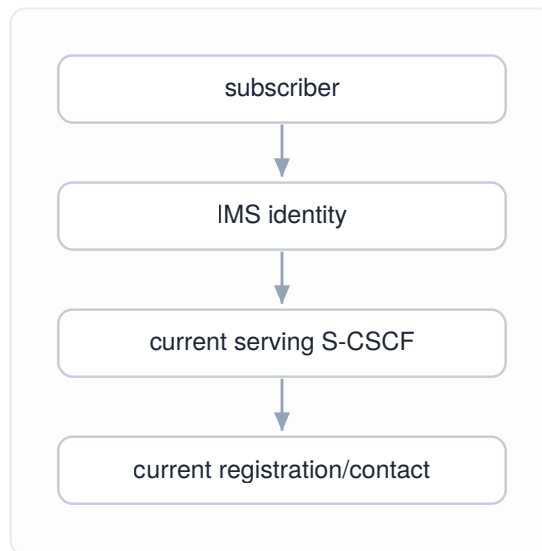


Figure 5.3 subscriber → current registration/contact

5.4 HSS/UDM enters the picture

In older IMS architectures:

Home Subscriber Server

was the central subscriber database associated with IMS.

It provides information required for:

- authentication
- S-CSCF assignment
- subscriber profile
- service configuration

In 5G deployments, subscriber data management uses the 5G Core's UDM/UDR architecture, while IMS still exists as the service layer.

This is why you will encounter both:

HLR
HSS
UDM
UDR

when studying different generations.

They're not four versions of exactly the same database.

5.5 S-CSCF

The S-CSCF is the central serving control function inside IMS.

A useful approximation:

- | | | |
|--------------|-----------------------|------------------------|
| • P-CSCF | • I-CSCF | • S-CSCF |
| • front door | • routing/entry point | • serving intelligence |

The S-CSCF handles session control and service logic interactions.

3GPP's IMS architecture explicitly describes the S-CSCF's role in service control and interaction with application servers. (3GPP)

5.6 IMS registration completes

Now Bob's phone may have state like:

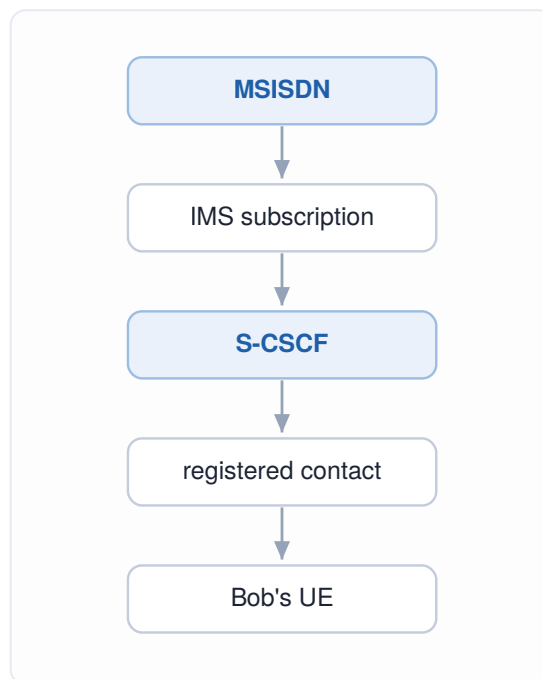


Figure 5.4 MSISDN → S-CSCF → Bob's UE

That is the key.

The telephone number itself doesn't directly contain:

- | | |
|--------------|-------------------|
| • gNB ID | • radio frequency |
| • IP address | • antenna |

The network has dynamically established the mapping.

CHAPTER 6

Placing the Call: the SIP INVITE

6.1 Now Alice presses CALL

6.2 SIP INVITE

6.3 P-CSCF receives the INVITE

6.4 S-CSCF analyzes the destination

6.1 Now Alice presses CALL

Finally.

Alice enters:

+91 98BBBBBB20

and presses:

CALL

The phone's telephony application asks the IMS stack to initiate a session.

Conceptually:

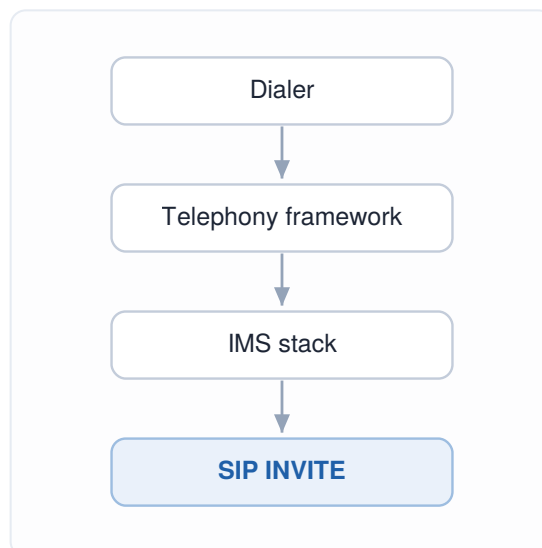


Figure 6.1 Dialer → Sip invite

The fundamental message is:

INVITE

Meaning approximately:

"I want to establish a multimedia session with this destination."

6.2 SIP INVITE

A highly simplified SIP message looks conceptually like:

INVITE sip:+9198BBBBBB20@operator.example SIP/2.0

From: Alice
To: Bob

Call-ID: abc123...

Contact: Alice@...

Content-Type: application/sdp

Real SIP messages contain substantially more information.

The body may contain SDP, Session Description Protocol.

SDP describes media capabilities such as:

- audio
- codec
- IP address
- port
- media parameters

So the INVITE isn't merely:

"call Bob"

It can effectively communicate:

"I want a session with Bob,

here are my media capabilities."

3GPP TS 24.229 specifies IMS call control using SIP and SDP. [3GPP Portal](#)

6.3 P-CSCF receives the INVITE

The first IMS hop:

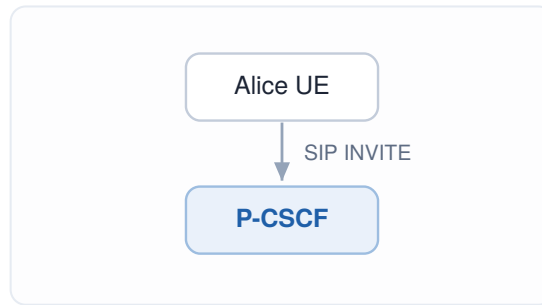


Figure 6.2 Alice UE → P-CSCF

The P-CSCF can perform functions such as:

- policy interaction
- security handling
- signaling forwarding
- QoS/resource authorization

3GPP specifications explicitly describe P-CSCF functions including resource authorization.

3GPP Portal

6.4 S-CSCF analyzes the destination

The INVITE eventually reaches Alice's S-CSCF.

Now:



Figure 6.3 S-CSCF → "Where does this number belong?"

This is where the telephone-number abstraction becomes routing intelligence.

The network may need to determine:

- country
- number range
- home network
- operator
- porting status
- IMS domain

Historically and in IMS procedures, destination analysis is followed by querying subscriber/location information and routing toward the appropriate serving CSCF. 3GPP's documented IMS call flow shows the originating S-CSCF analyzing the destination, the I-CSCF querying the HSS for current location/serving-CSCF information, and the request then being forwarded to the destination S-CSCF. (3GPP)

Inter-Operator Routing

7.1 If Bob is on another operator

7.2 I-CSCF

7.3 Bob's S-CSCF

7.4 TAS / Application Server

7.1 If Bob is on another operator

Then:

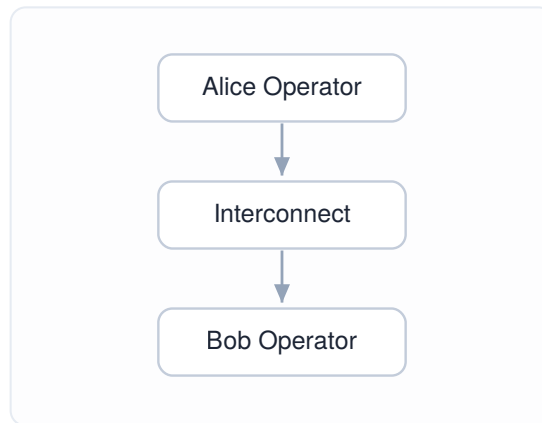


Figure 7.1 Alice Operator → Bob Operator

The IMS signaling can traverse an inter-operator network.

Conceptually:

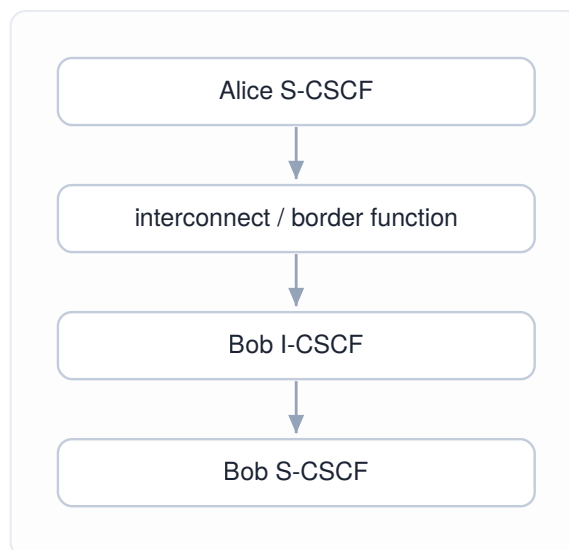


Figure 7.2 Alice S-CSCF → Bob S-CSCF

The exact architecture varies.

But the key principle remains:

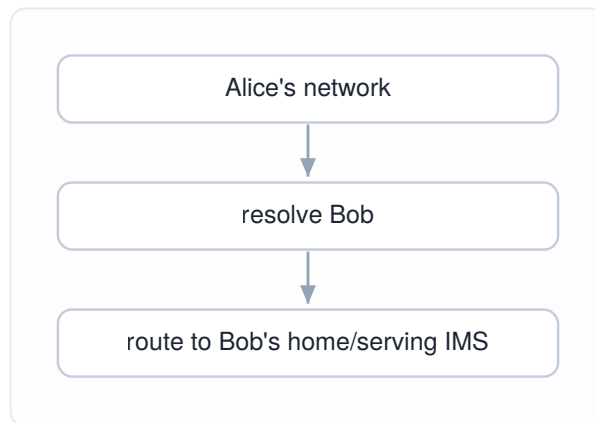


Figure 7.3 Alice's network → route to Bob's home/serving IMS

7.2 I-CSCF

Bob's operator may use an:

I-CSCF

Interrogating-CSCF.

A useful mental model:

I-CSCF

IMS ingress/routing function

It can query subscriber information to determine which S-CSCF is currently serving the destination subscriber.

The historical 3GPP IMS procedure explicitly describes:

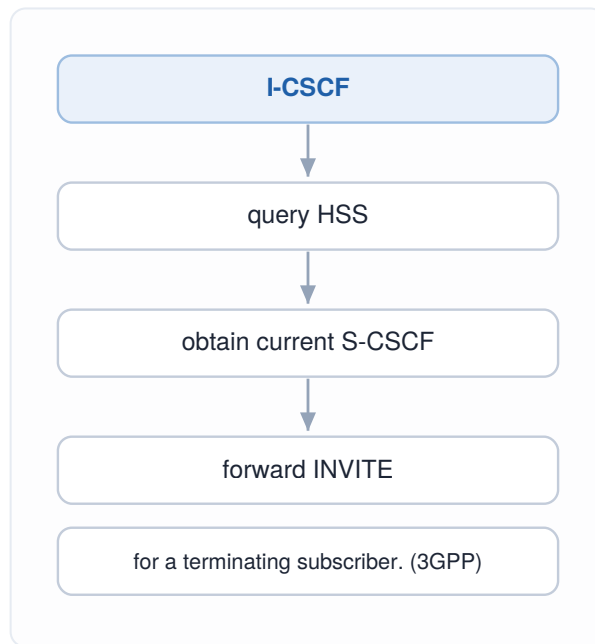


Figure 7.4 I-CSCF → obtain current S-CSCF → for a terminating subscriber. (3GPP)

7.3 Bob's S-CSCF

Now:

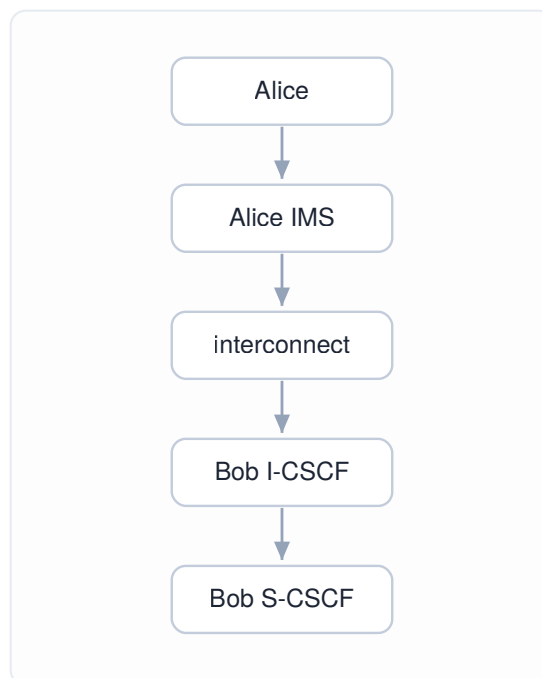


Figure 7.5 Alice → interconnect → Bob S-CSCF

The S-CSCF can apply Bob's service logic.

For example:

Is Bob registered?

Does he have call forwarding?

Does he have barring?

Is voicemail configured?

Is this an enterprise service?

This is where service logic enters the call.

7.4 TAS / Application Server

Telecommunications networks often use an:

Application Server

or specialized:

Telephony Application Server

TAS

for supplementary services.

Examples:

- call forwarding
- call waiting
- conference calling
- number translation
- voicemail integration
- enterprise PBX logic

So the call isn't simply:

number → phone

It can become:

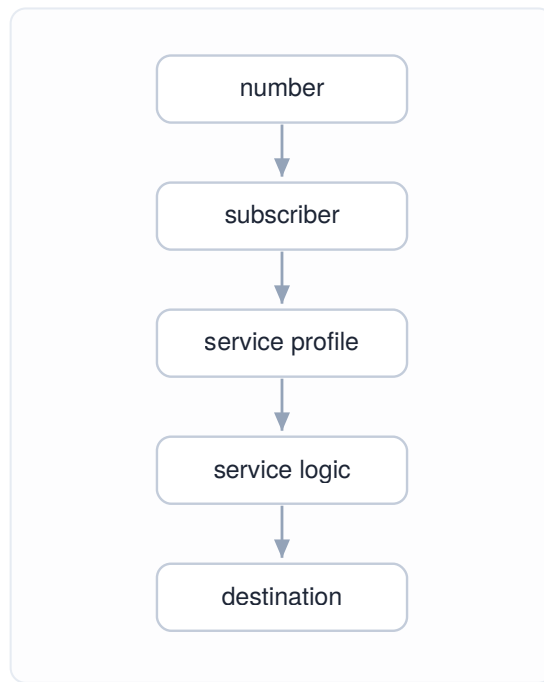


Figure 7.6 number → service profile → destination

IMS was deliberately designed to support application servers and service-control logic. (3GPP)

Finding Bob: Paging and Alerting

8.1 Bob is currently idle

8.2 Paging is not "call every tower"

8.3 Bob's phone responds

8.4 SIP alerting

8.5 Bob answers

8.1 Bob is currently idle

Suppose Bob's phone isn't actively transmitting user-plane traffic.

His device may still be:

- registered
- reachable
- idle

Now the network has to get his attention.

This is where paging enters.

Conceptually:

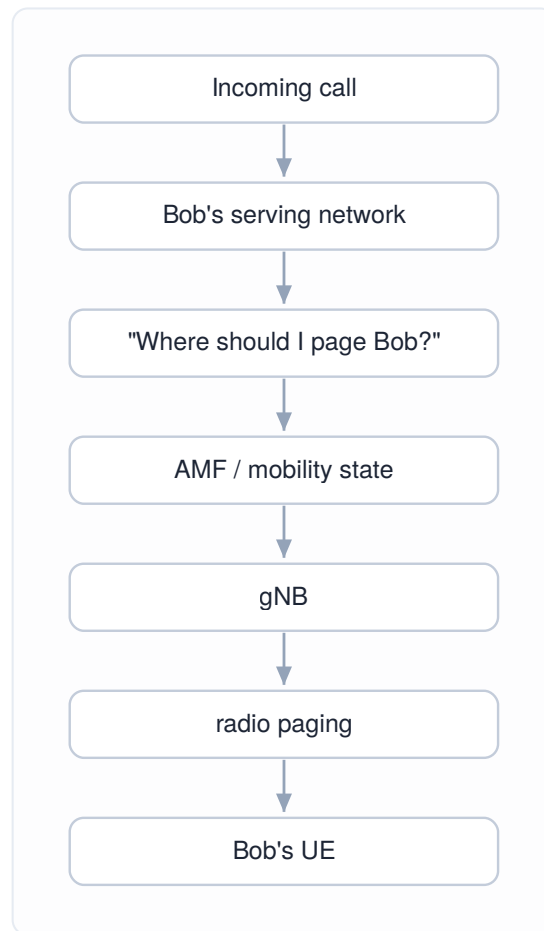


Figure 8.1 Incoming call → AMF / mobility state → Bob's UE

The exact paging procedure is specified as part of the 5G System procedures in TS 23.502.

8.2 Paging is not "call every tower"

This matters.

The network has mobility/location information.

Suppose Bob is known to be in a particular tracking area:

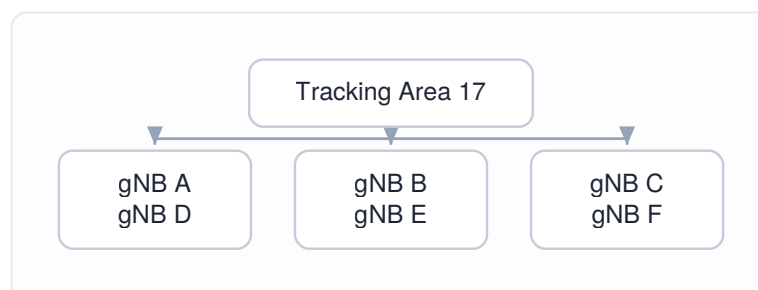


Figure 8.2 Tracking Area 17 → gNB A / gNB B / gNB C

The network can page within the appropriate area rather than blindly broadcasting globally.

Conceptually:

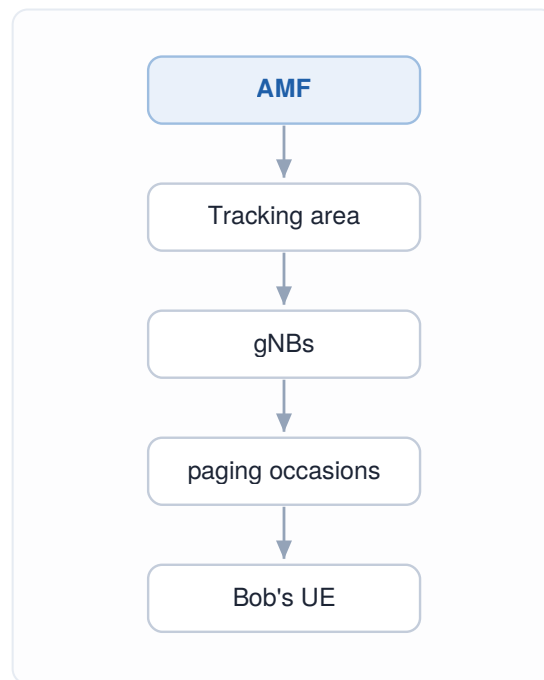


Figure 8.3 AMF → gNBs → Bob's UE

The UE periodically monitors specific paging opportunities to conserve battery.

Otherwise your phone would need to listen continuously.

Battery technology would have had to evolve into something resembling a brick with a USB cable permanently attached.

8.3 Bob's phone responds

Bob's UE detects the paging.

It effectively says:

"I am here."

The network then establishes or resumes the required signaling connectivity.

Now the call setup can continue.

8.4 SIP alerting

Bob's IMS stack receives the INVITE.

His phone starts ringing.

The signaling can produce responses such as:

100 Trying
180 Ringing
200 OK

Very roughly:

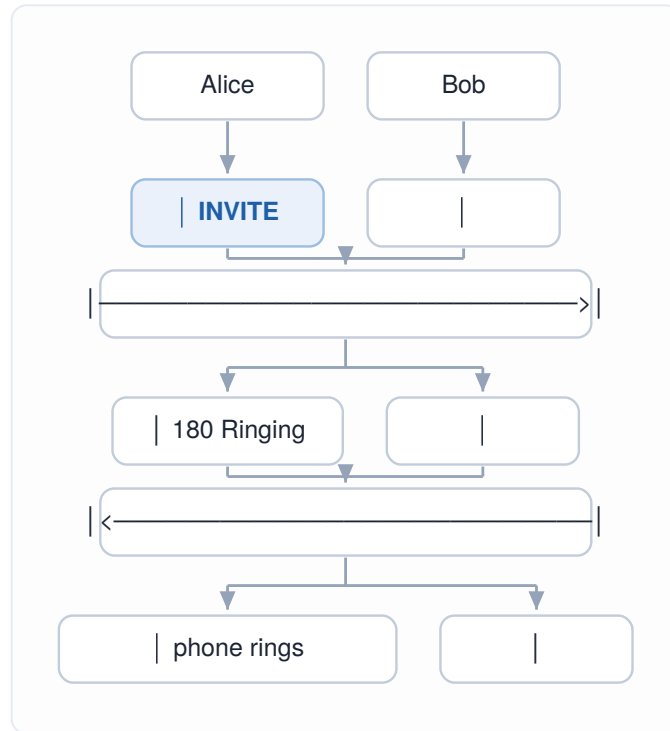


Figure 8.4 Alice / Bob → | 180 Ringing / | → | phone rings / |

At this stage:

call requested

call answered

8.5 Bob answers

Bob presses:

ANSWER

Now his UE generates the successful SIP response:

200 OK

Conceptually:

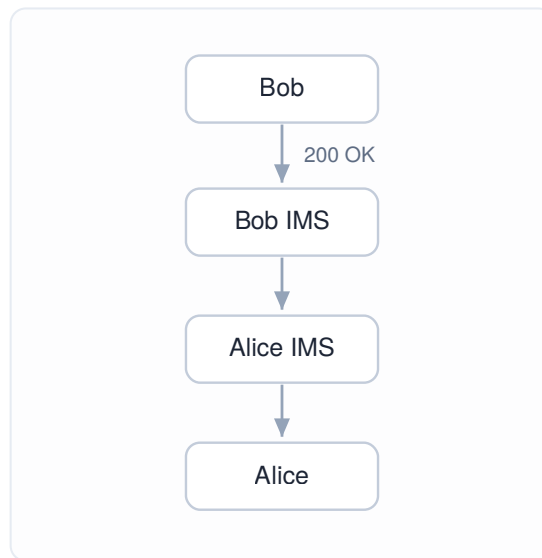


Figure 8.5 Bob → Alice

3GPP's IMS call-flow documentation explicitly describes the final 200 OK being returned over the signaling path when the called party accepts the incoming call. (3GPP)

Media: Codecs, QoS and RTP

9.1 But we still don't have the conversation

9.2 Codec negotiation

9.3 Now QoS becomes important

9.4 The voice packet path

9.5 RTP

9.6 One second of speech

9.1 But we still don't have the conversation

This is the subtle part.

We've established:

CALL CONTROL

but we haven't necessarily established the optimized voice bearer yet.

The signaling says approximately:

- "These are our media capabilities."
- "These are our media capabilities."
- "Okay, let's use this media configuration."

The SDP information negotiates the media parameters.

9.2 Codec negotiation

Suppose Alice supports:

AMR
AMR-WB
EVS

and Bob supports:

AMR-WB
EVS

The endpoints/network negotiate an appropriate codec.

Modern mobile voice can use codecs such as:

AMR
AMR-WB
EVS

The codec converts:

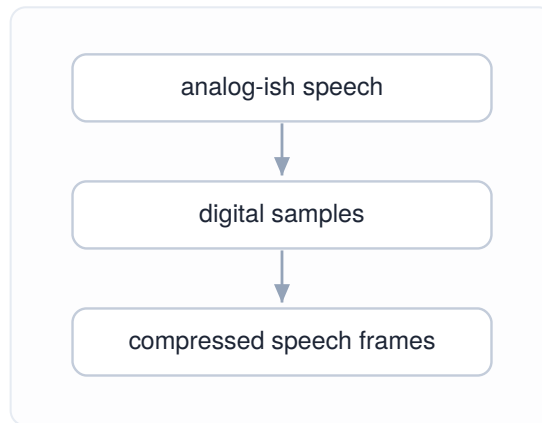


Figure 9.1 analog-ish speech → compressed speech frames

For example:

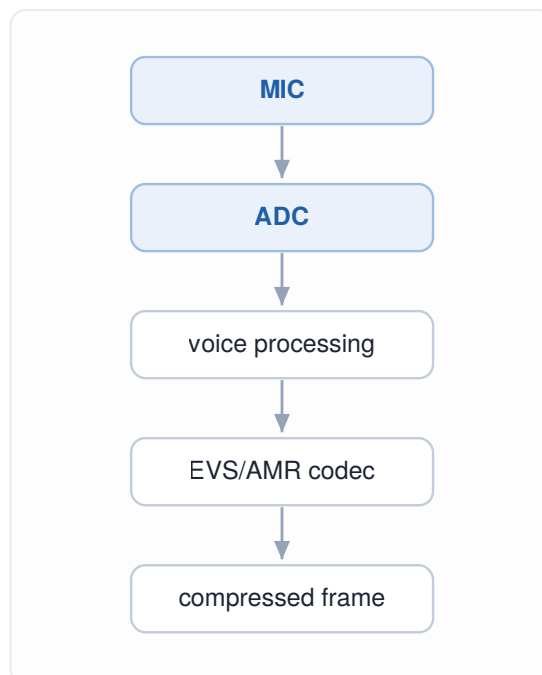


Figure 9.2 MIC → voice processing → compressed frame

9.3 Now QoS becomes important

Voice is not ordinary best-effort Internet traffic.

You don't want:

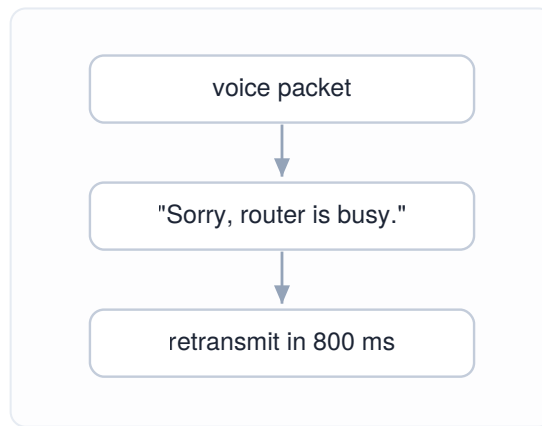


Figure 9.3 voice packet → retransmit in 800 ms

because conversation would become interpretive dance.

The network therefore establishes appropriate QoS treatment.

In 5G, the relevant abstraction is a:

QoS Flow

A voice flow can receive differentiated treatment from ordinary data traffic.

The architecture includes policy and charging functions such as:

PCF
SMF
UPF

to coordinate policy and packet handling.

3GPP maintains dedicated specifications for 5G procedures and QoS architecture.

9.4 The voice packet path

Now we can finally send actual speech.

Alice says:

"Hello."

Her microphone converts the acoustic waveform into an electrical signal.

Then:

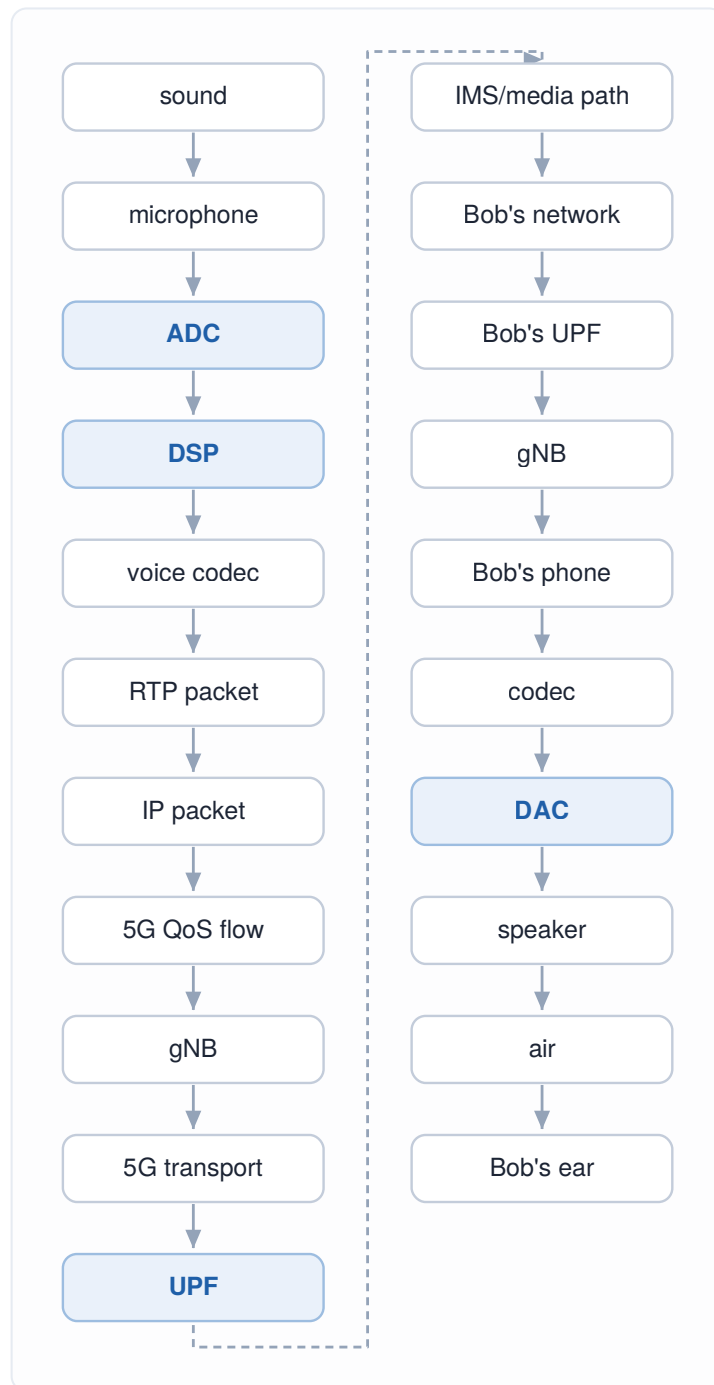


Figure 9.4 sound → UPF → Bob's ear

That's the entire journey of one tiny chunk of speech.

9.5 RTP

The actual media is commonly carried using:

RTP, Real-time Transport Protocol

Conceptually:

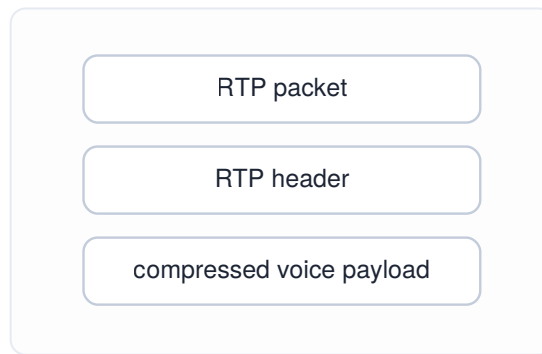


Figure 9.5 RTP packet → compressed voice payload

The payload contains encoded audio.

RTP provides sequencing/timing information useful for real-time media.

And importantly:

- SIP
- call/session signaling
- RTP
- media

Don't conflate them.

9.6 One second of speech

Imagine Alice says:

"hello"

Her voice becomes a waveform:

- /\
- /\ /\
- ____/____/____

The microphone samples it.

The codec processes those samples.

Then packets are created:

- Packet 1
- Packet 2
- Packet 3
- Packet 4

Each packet contains a piece of the encoded speech.

The network transports them independently.

So the conversation isn't:

ONE CONTINUOUS ELECTRICAL WIRE

anymore.

It is more like:

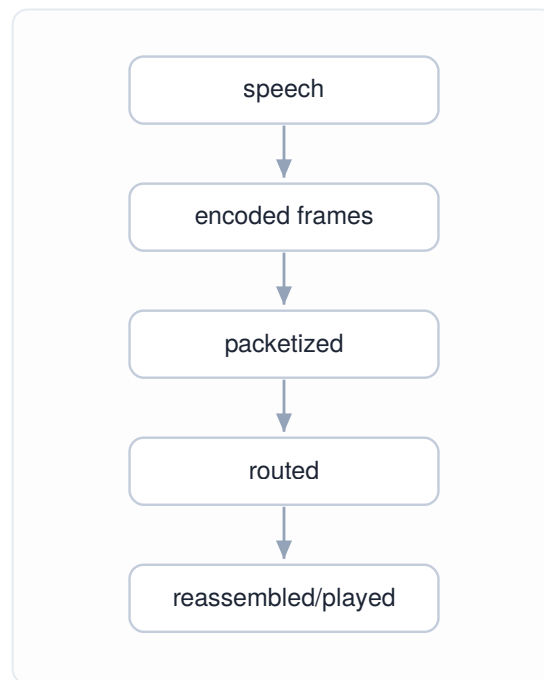


Figure 9.6 speech → packetized → reassembled/played

The 5G Radio Stack

10.1 The actual 5G radio stack

10.2 PHY

10.3 MAC

10.4 RLC

10.5 PDCP

10.6 gNB

10.7 N2 and N3

10.8 UPF

10.1 The actual 5G radio stack

Now let's descend another level.

Alice's UE communicates with the gNB using the 5G NR stack.

Simplified:

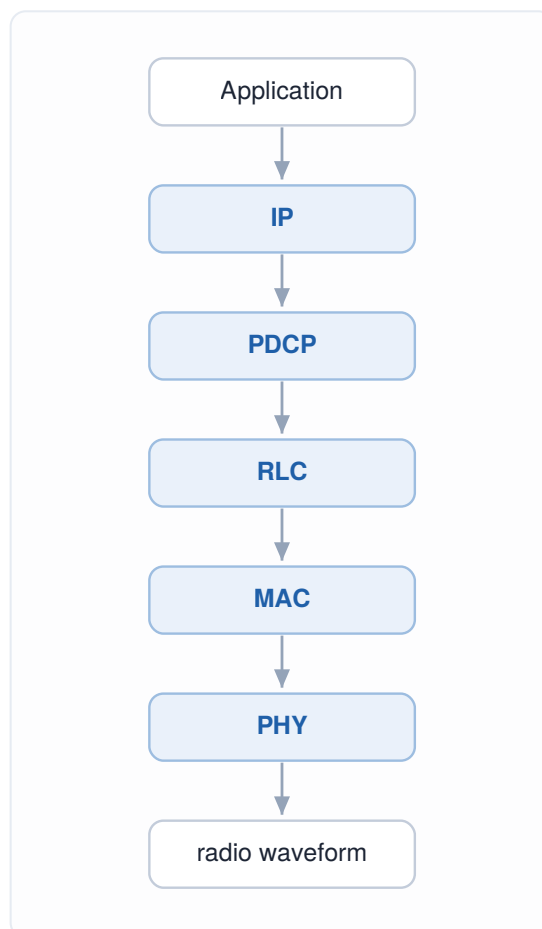


Figure 10.1 Application → RLC → radio waveform

Each layer has a different responsibility.

10.2 PHY

The physical layer deals with actual radio transmission.

Conceptually:

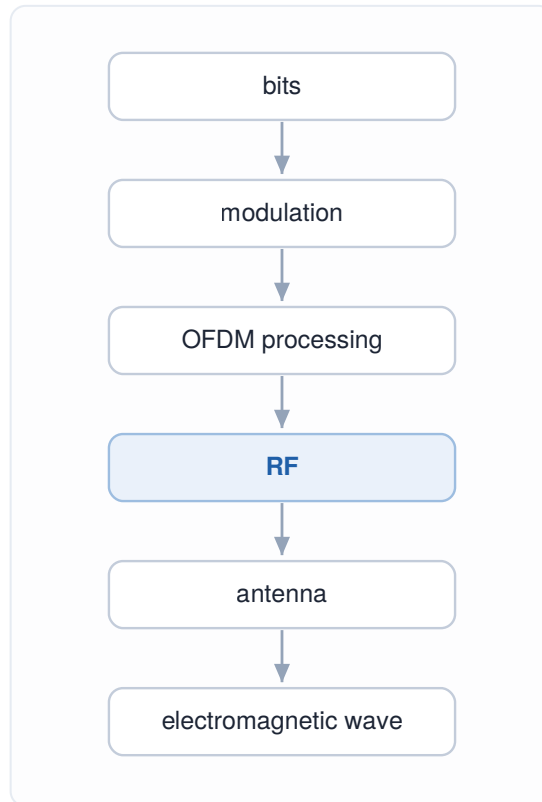


Figure 10.2 bits → RF → electromagnetic wave

The phone converts digital information into radio-frequency signals.

The gNB reverses the process.

10.3 MAC

MAC handles things including:

- radio resource allocation
- scheduling
- multiplexing
- HARQ-related mechanisms

The scheduler determines who gets radio resources.

For example:

UE A → resource blocks

UE B → resource blocks

UE C → resource blocks

This happens dynamically.

10.4 RLC

RLC provides functions related to:

- segmentation
- reassembly
- retransmission modes

depending on configuration.

10.5 PDCP

PDCP handles functions such as:

- header compression
- integrity protection
- reordering
- ciphering
- sequence numbering

depending on the traffic/bearer and configuration.

So the packet passes through several layers before becoming a radio transmission.

10.6 gNB

The gNB is the 5G base station.

It is not simply:

"5G antenna"

It is a substantial computing/networking system.

It handles:

- PHY
- MAC
- RLC
- PDCP
- RRC
- NG interface
- scheduling
- radio resource management
- mobility-related procedures

The gNB connects toward the 5G Core.

10.7 N2 and N3

The 5G architecture separates important interfaces.

Very simplified:

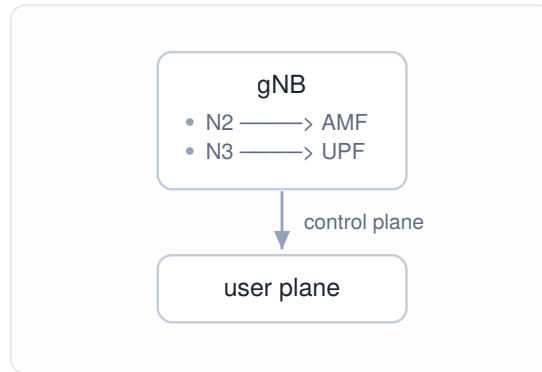


Figure 10.3 gNB → user plane

This is an extremely useful architecture distinction.

- | | |
|---------------------|----------------------|
| • N2 | • N3 |
| • control signaling | • user-plane traffic |

So the actual voice packets do not necessarily go:

gNB → AMF → voice

The AMF is control-plane machinery.

User traffic is handled through:

gNB → UPF

10.8 UPF

The User Plane Function is basically where packet forwarding happens inside the 5G Core.

Conceptually:

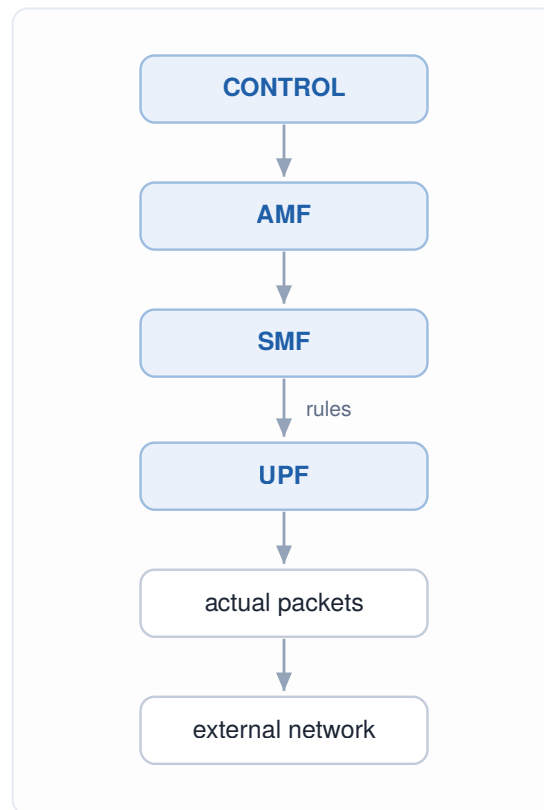


Figure 10.4 CONTROL → UPF → external network

The SMF tells the UPF what treatment/forwarding rules apply.

Then the UPF handles user-plane packets at high throughput.

This is architecturally analogous to separating:

control logic

from:

fast packet forwarding

which should look familiar if you've studied networking.

Generations Compared

11.1 What happened to the old MSC?

11.2 LTE VoLTE

11.3 5G VoNR

11.4 What if the destination is a landline?

11.5 What happens when Alice hangs up?

11.1 What happened to the old MSC?

This is where generations become confusing.

2G/3G

You might have:

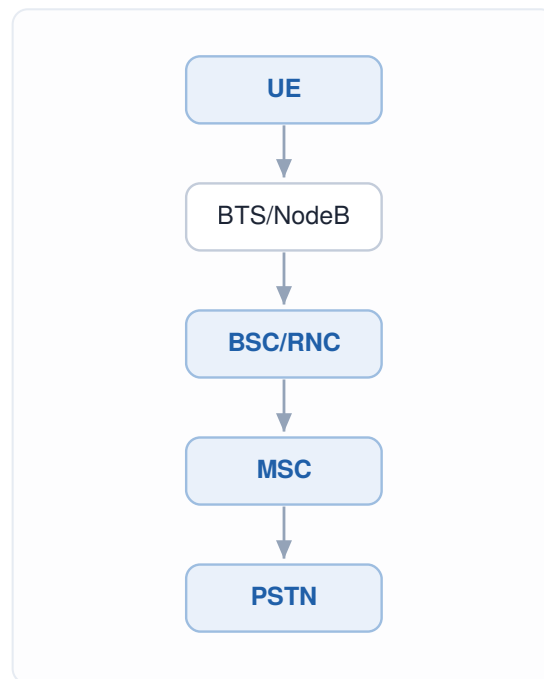


Figure 11.1 UE → BSC/RNC → PSTN

The MSC was central to circuit-switched voice.

4G

The architecture becomes packet-oriented:

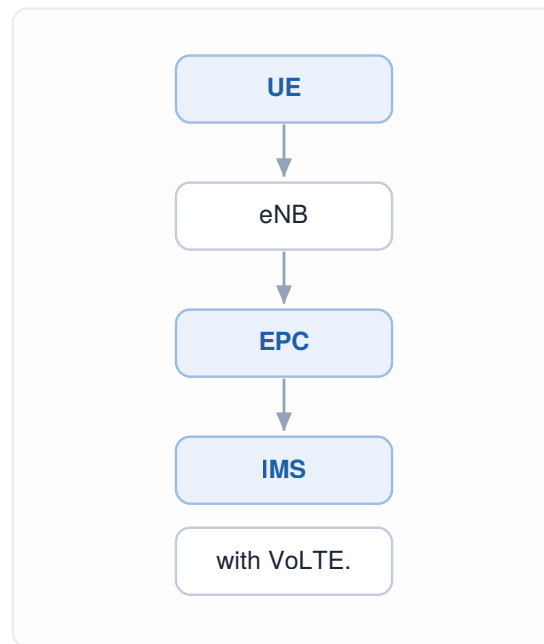


Figure 11.2 UE → EPC → with VoLTE

5G SA

It becomes:

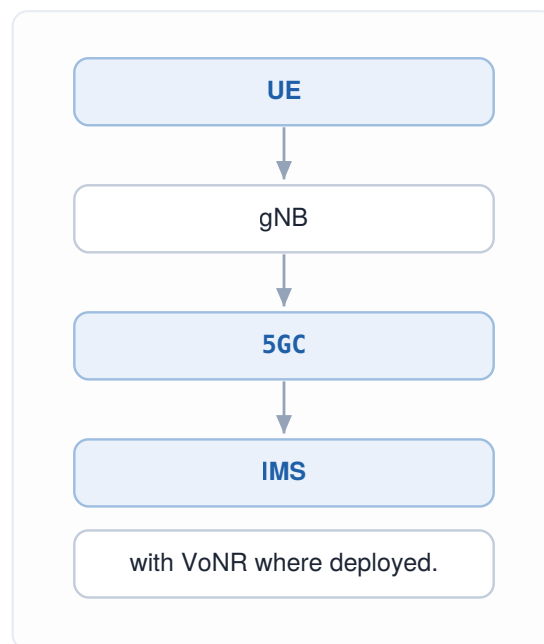


Figure 11.3 UE → 5GC → with VoNR where deployed

MSC

didn't simply become:

AMF

Those functions are architecturally different.

That's an important exam/interview trap.

11.2 LTE VoLTE

Now compare the 5G SA architecture.

VoLTE:

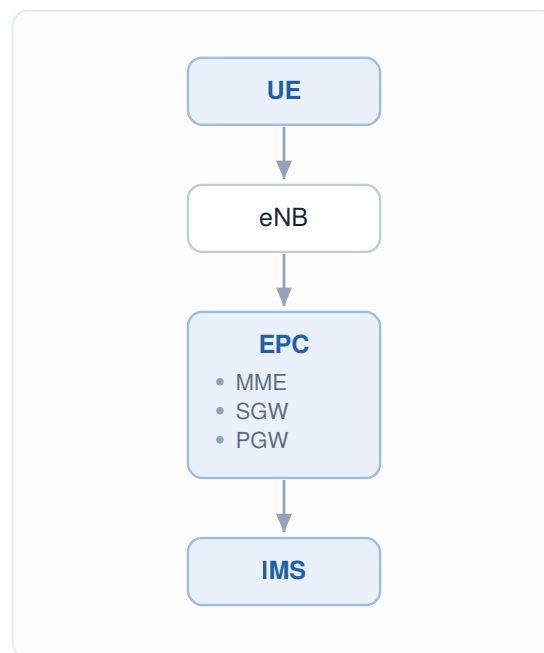


Figure 11.4 UE → IMS

Voice is carried as an IP service through IMS.

- 4G radio
- EPC packet core
- IMS
- VoLTE

11.3 5G VoNR

For 5G standalone:

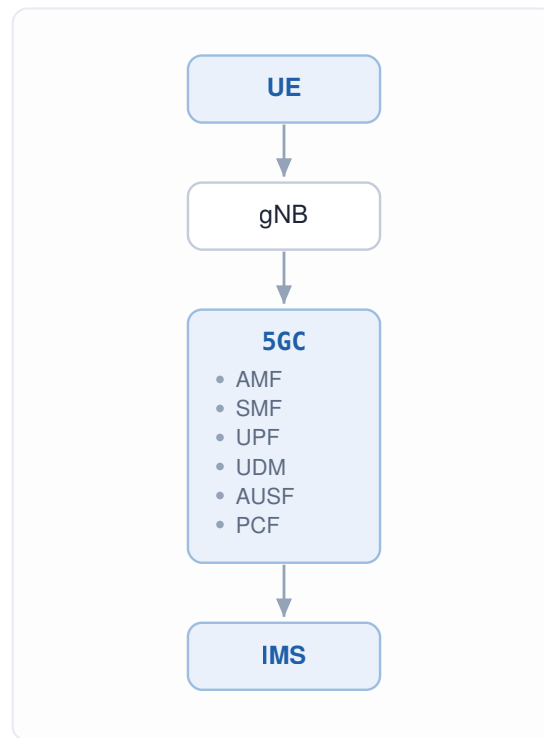


Figure 11.5 UE → IMS

Then:

- 5G NR
- 5G Core
- IMS
- VoNR

3GPP's current TS 23.502 is the central specification for 5GS procedures and continues to evolve through current releases.

11.4 What if the destination is a landline?

Then the architecture may eventually need interworking:

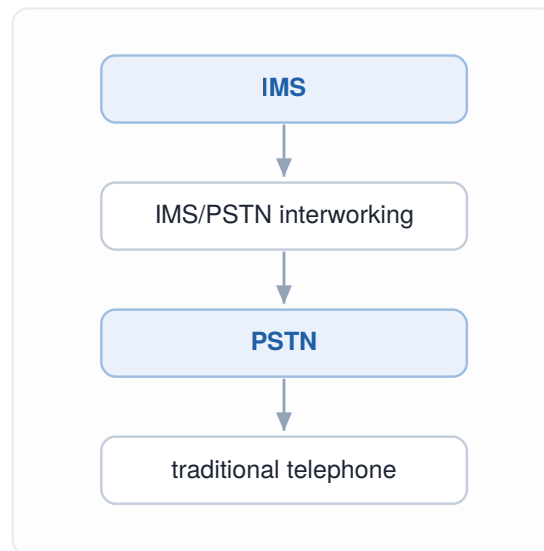


Figure 11.6 IMS → traditional telephone

3GPP maintains a dedicated specification for interworking between the IMS core and circuit-switched networks, TS 29.163. [3GPP Portal](#)

So a modern smartphone can communicate with an ancient-looking landline because the network contains translation/interworking machinery.

This is one of the great achievements of telecommunications engineering:

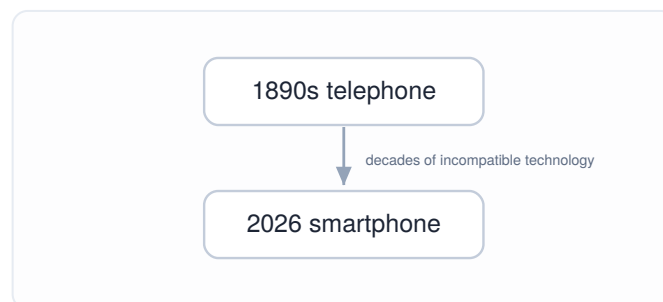


Figure 11.7 1890s telephone → 2026 smartphone

and they can still talk.

11.5 What happens when Alice hangs up?

Alice presses:

END

The UE sends signaling indicating that the session should terminate.

Conceptually:

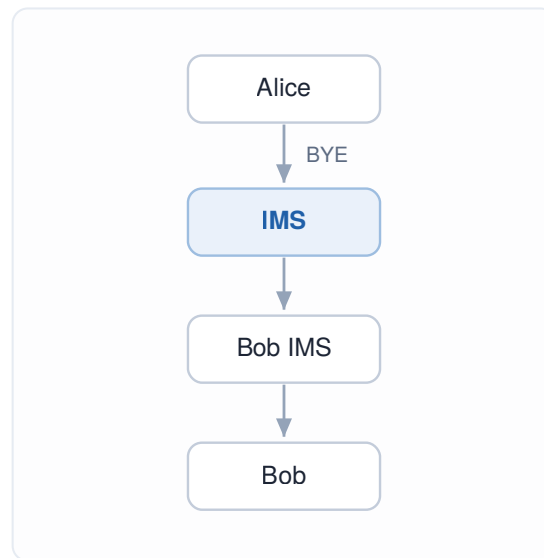


Figure 11.8 Alice → Bob

Then:

- RTP stops
- QoS resources released
- PDU/QoS state updated
- session state cleared
- billing records finalized

The network tears down the state associated with the call.

The Entire Call in One Sequence

12.1 The entire call in one sequence

12.1 The entire call in one sequence

Now compress everything:

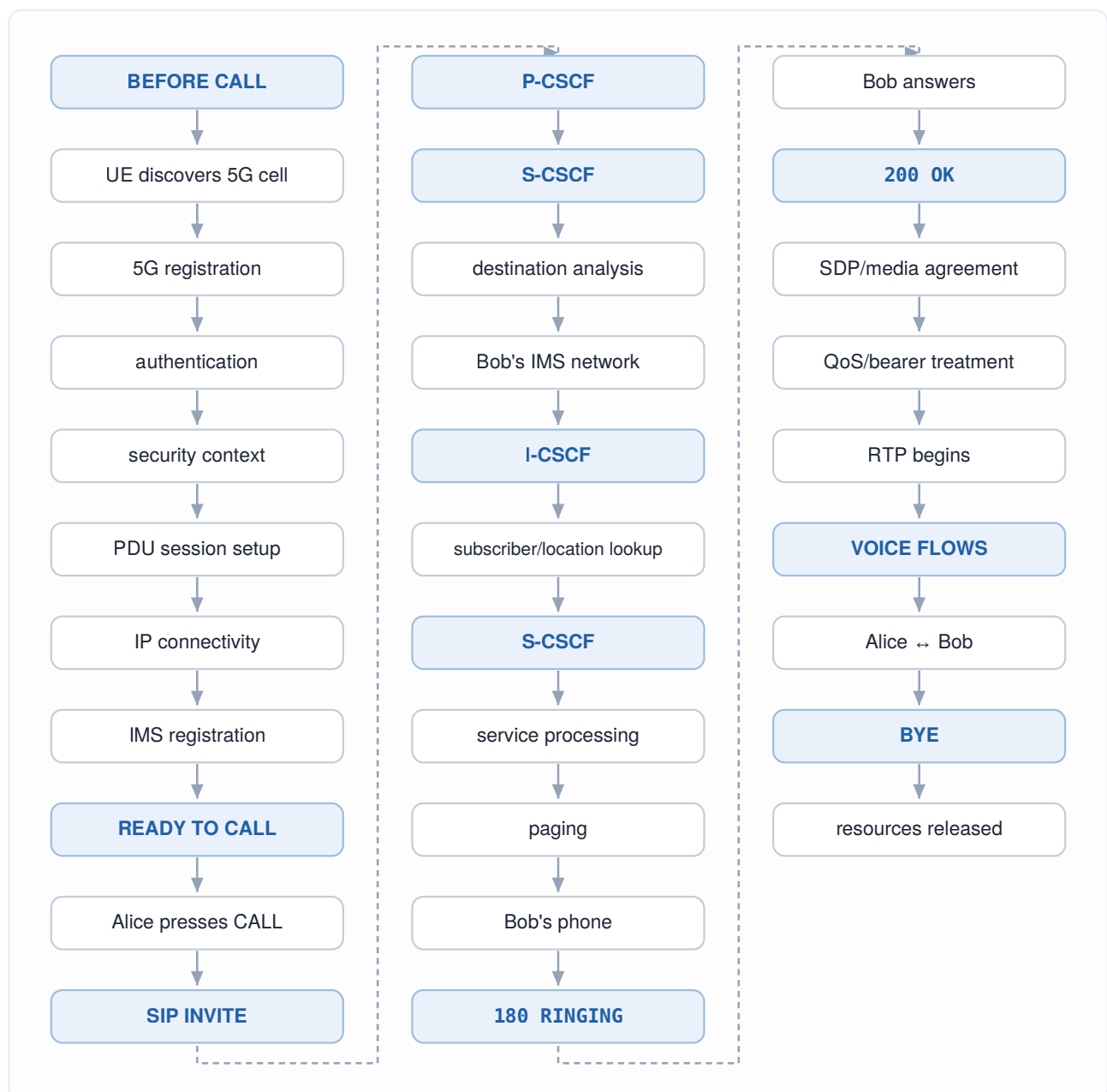


Figure 12.1 Before call → I-CSCF → resources released

That's the complete conceptual lifecycle.

Identifiers, Mobility and Roaming

13.1 The really interesting part: there are multiple identifiers

13.2 This is exactly like a computer network

13.3 Why the number doesn't need to know your location

13.4 And now you can understand roaming

13.5 Where SS7 fits now

13.1 The really interesting part: there are multiple identifiers

During this one call, the network is dealing with a ridiculous number of identities.

Something conceptually like:

Alice:

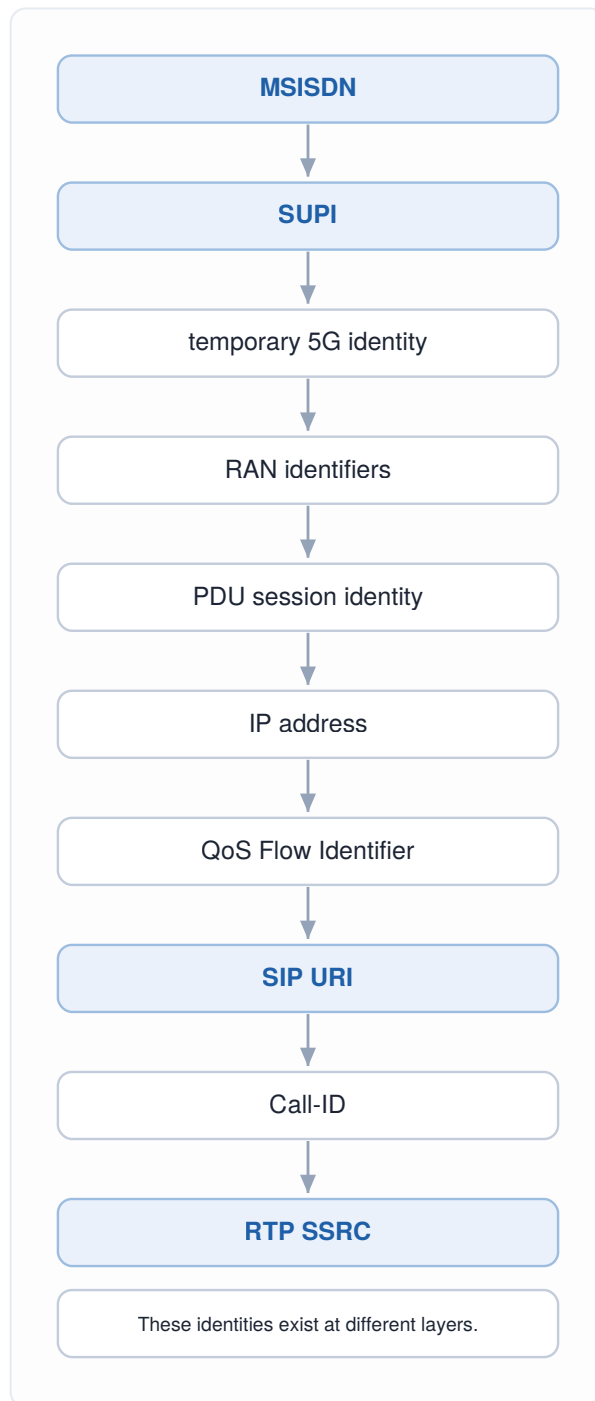


Figure 13.1 MSISDN → IP address → These identities exist at different layers

For example:

+91 98AAAAAA10

might identify Alice's public telephony identity.

A SIP URI identifies her at the IMS layer.

An IP address identifies an endpoint at the IP layer.

A QoS Flow Identifier identifies a 5G QoS flow.

An RTP SSRC identifies a media source within RTP.

None of these are "the phone's address."

They're different abstractions.

13.2 This is exactly like a computer network

You can map the concepts roughly like this:

TELECOM	COMPUTER NETWORKING ANALOGY
MSISDN	Application-level logical identifier
IMS identity	Service identity
IP address	Network-layer endpoint
GTP-U	Tunnel mechanism
QoS Flow	Traffic class/flow
RTP	Application media transport
SIP	Session-control protocol
gNB	Access network device
UPF	User-plane forwarding
AMF	Control/mobility function
DNS-like resolution	Number/service routing
Paging	Reachability mechanism

Table 13.1

The analogy isn't exact, but it's extremely useful.

13.3 Why the number doesn't need to know your location

This should now be obvious.

Suppose Bob moves:



Figure 13.2 Delhi → London

His:

MSISDN

can remain constant.

Meanwhile:

- serving gNB
- tracking area
- AMF context
- IP/session state
- IMS contact

can change.

Therefore:

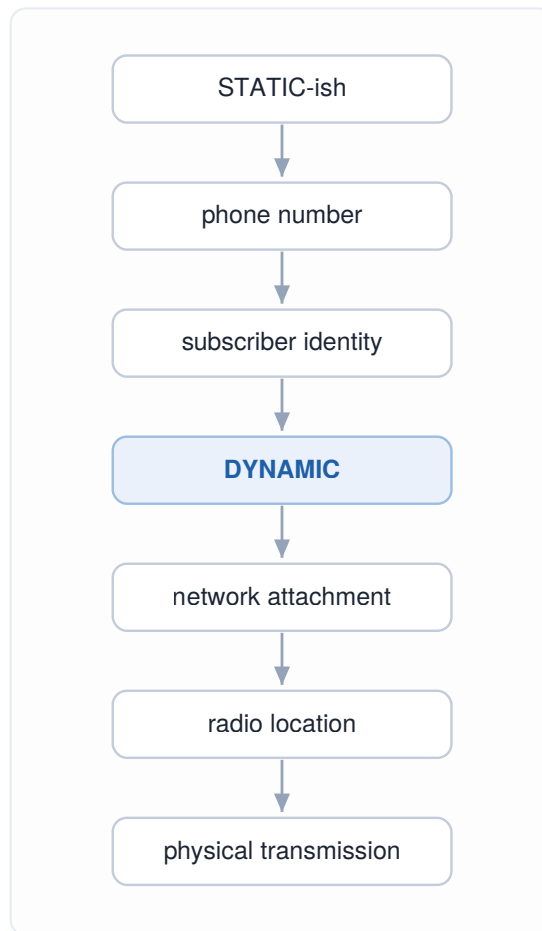


Figure 13.3 STATIC-ish → DYNAMIC → physical transmission

That separation is the fundamental reason mobile telecommunications works.

13.4 And now you can understand roaming

Suppose Bob takes his Indian SIM to Germany.

His number remains:

+91 ...

but the radio network is now German.

Conceptually:

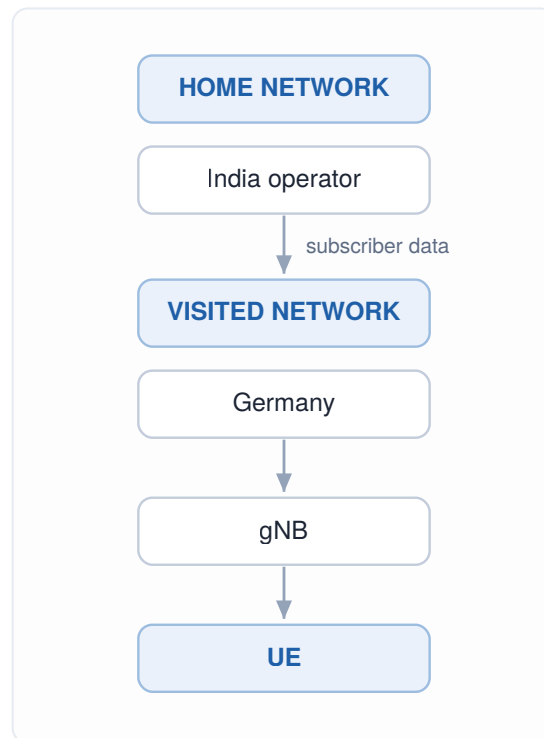


Figure 13.4 Home network → Germany → UE

The visited network provides access while the home network remains authoritative for the subscriber relationship.

That is another demonstration that:

- subscriber identity
- radio access identity
- physical location

13.5 Where SS7 fits now

This is important because you mentioned:

- SS7
- Diameter
- SIP

They belong to different generations/layers.

A rough historical progression:

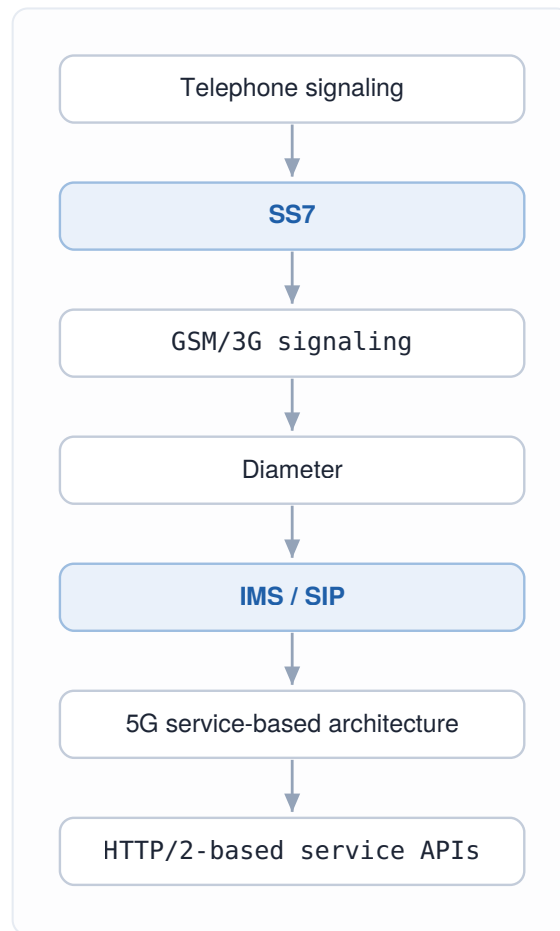


Figure 13.5 Telephone signaling → Diameter → HTTP/2-based service APIs

This isn't a strict replacement chain.

Real networks are ugly because legacy infrastructure doesn't politely disappear when a new standard arrives.

You therefore get:

- | | | |
|------------|----------|--------|
| • SS7 | • HTTP/2 | • PFCP |
| • Diameter | • GTP | • RTP |
| • SIP | • NGAP | |

all potentially participating in different portions of a modern telecom system.

Protocol Roles and the Deepest Model

14.1 What each protocol is doing

14.2 The deepest mental model

14.3 The entire thing from digit to electromagnetic wave

14.1 What each protocol is doing

This table is worth memorizing:

PROTOCOL	MAIN JOB
SIP	Establish/control multimedia sessions
SDP	Describe media/session parameters
RTP	Carry real-time media
RTCP	RTP control/quality feedback
HTTP/2	5G Service-Based Architecture communication
NGAP	gNB ↔ 5GC control signaling
NAS	UE ↔ 5GC mobility/session signaling
GTP-U	User-plane tunneling
PFCP	SMF ↔ UPF control
Diameter	Legacy/4G/IMS-related AAA/subscriber/policy functions
SS7/MAP	Legacy cellular signaling
RRC	UE ↔ gNB radio control
PDCCP/RLC/MAC/PHY	Radio protocol stack

Table 14.1

This is where telecommunications starts looking less like "a phone call" and more like what it really is:

A distributed state machine implemented across several protocol stacks.

14.2 The deepest mental model

Don't visualize a phone call as:

PHONE ————— PHONE

That's the primitive mental model.

Visualize it as:

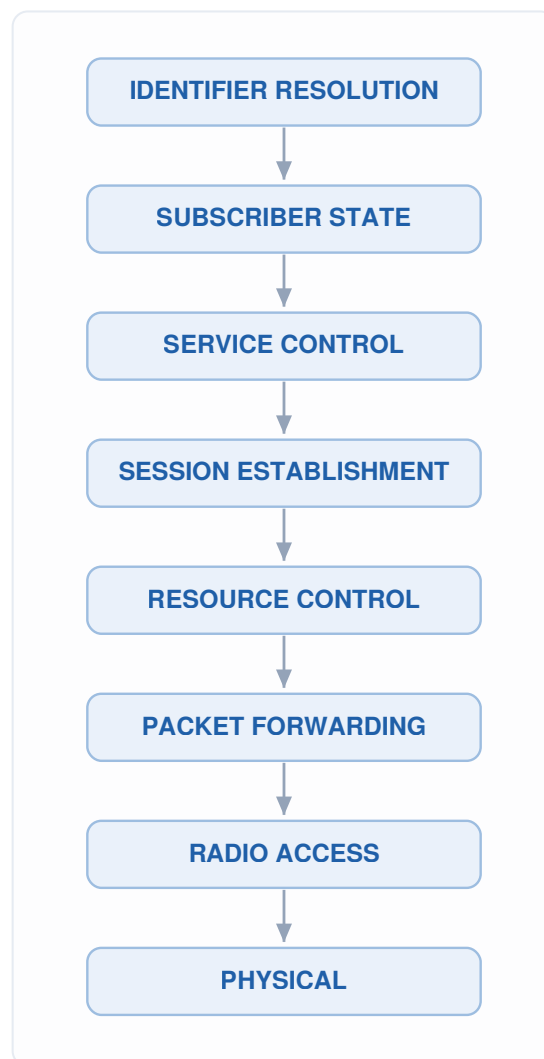


Figure 14.1 Identifier resolution → Resource control → PHYSICAL

And simultaneously:

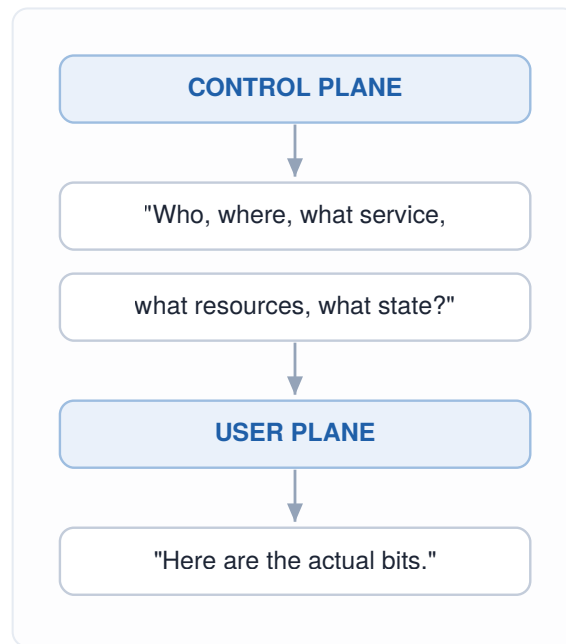


Figure 14.2 Control plane → what resources, what state?" → "Here are the actual bits."

That is the architecture.

14.3 The entire thing from digit to electromagnetic wave

If we descend all the way:

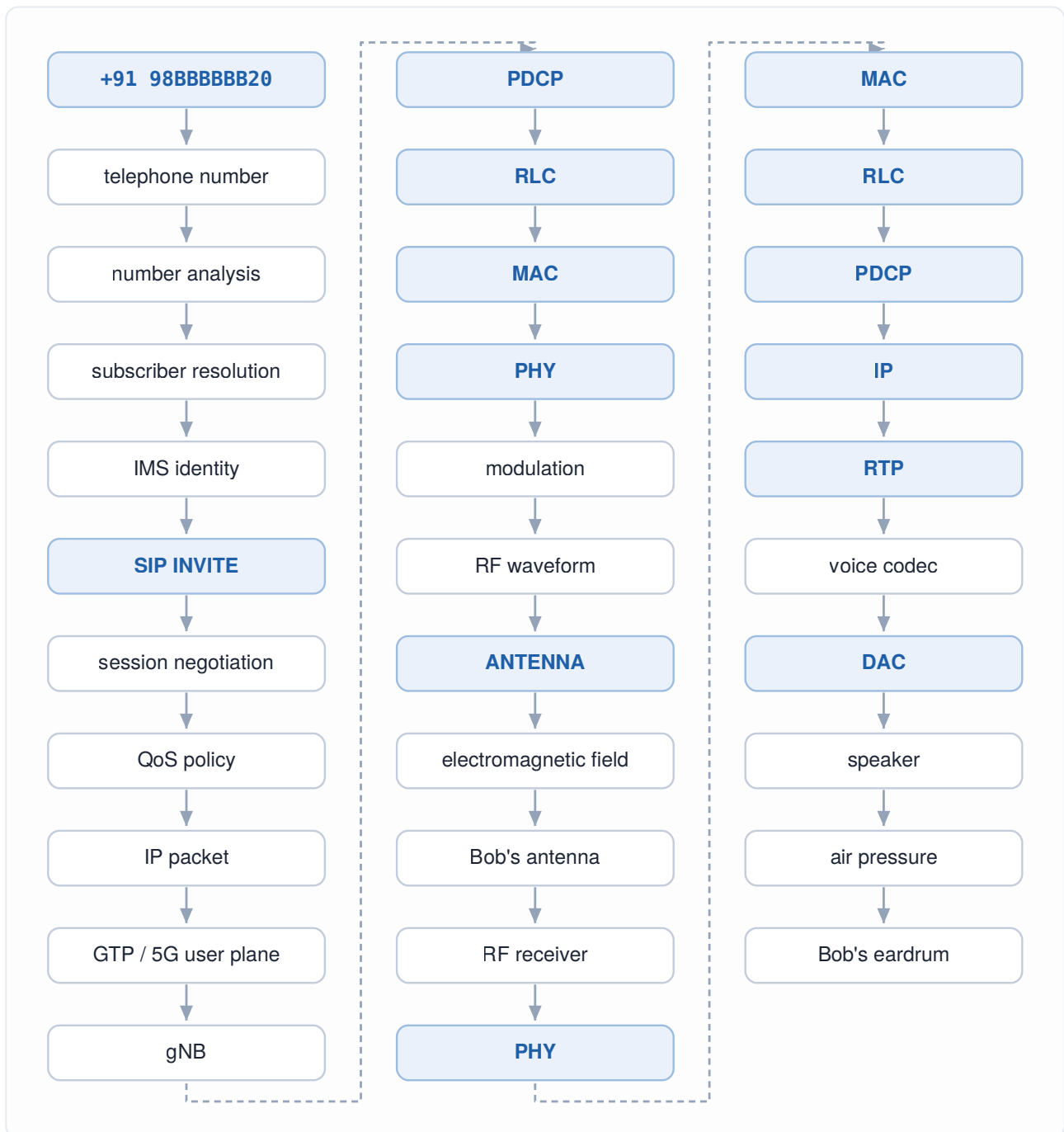


Figure 14.3 +91 98bbbbbb20 → RF waveform → Bob's eardrum

That is the full conceptual descent from a human-readable number to a pressure wave in another person's ear.

And that is why learning telecommunications properly is useful for a systems-oriented CS student: it forces you to see that identifiers, control state, routing, resource allocation, transport, and physical transmission are separate abstractions that cooperate without being the same thing.

The next layer below this is even nastier: we can take one actual SIP INVITE, one 5G NAS message, one NGAP message, one GTP-U packet and one RTP packet and dissect their headers field-by-field, showing exactly what bytes exist on the wire and how the packet

changes as it traverses $UE \rightarrow gNB \rightarrow 5GC \rightarrow IMS \rightarrow \text{destination}$. That is the point where this stops being architecture theory and becomes packet-level telecommunications engineering.